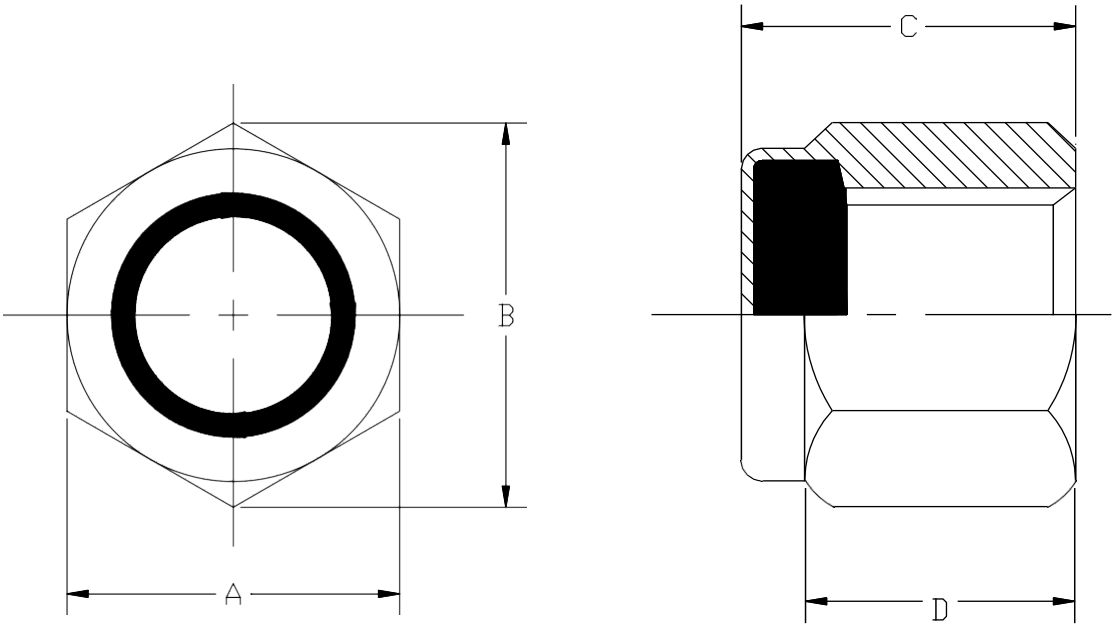
INCH-POUND
MIL-DTL-45913/3C
w/ AMENDMENT 2
20 December 2024
SUPERSEDING
MIL-DTL-45913/3C
w/ AMENDMENT 1
21 February 2024

DETAIL SPECIFICATION SHEET

NUT, SELF-LOCKING, HEXAGON, NON-METALLIC LOCKING FEATURE, 250 °F, UNC/UNF-3B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-45913.



(All dimensions are in inches.)

FIGURE 1. Nut, self-locking, hex, non-metallic locking feature UNC/UNF-3B.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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REQUIREMENTS:

MATERIAL:

Steel and alloy steel shall be in accordance with SAE J995 grade 5 or 8, as specified in part-number. Corrosion resisting steel (stainless steel) shall be either (a) Grade 303 or any Group 1 alloy in ASTM F594, or (b) Grade 316. Corrosion resistant steel shall have a minimum ultimate tensile strength of 70 KSI and be non-magnetic.
Copper alloy (brass) shall be in accordance with ASTM B16/B16M or ASTM B21/B21M in the half-hard temper (H02).
Aluminum alloy shall be 2024 in accordance with SAE AMS4120 or SAE AMS4339 and have a minimum ultimate tensile strength of 62 KSI.

PLATING/FINISH:

See Table III.

SURFACE TEXTURE. Shall be Ra 125 microinches in accordance with ASME B46.1.

THREADS. Shall be coarse or fine threaded, class 3B, as specified in part-number, in accordance with ASME B1.1.

WORKMANSHIP. Parts shall be manufactured and processed with a level of care and workmanship befitting this type of product intended for general purpose military application. Parts shall be free from dirt, grease, loose or deposited foreign materials, sharp edges, burrs, chips, physical malformations or anything that might prevent the part from reliably performing its intended function.

PART NUMBER. The part number shall consist of the basic number of this specification sheet followed by a dash, a dash number from Table I, a material code from Table II, and a finish/plate code (when required) from Table III.

EXAMPLE OF PART NUMBER:

M45913/3- 4C G5 QX

- QX — Finish Code (see Table III)
- G5 — Material Code (see Table II)
- 4C — Dash number from Table I to indicate size and thread.
- M45913/3- — "M" plus the Basic Specification Sheet Number followed by "-"

M45913/3-4CG5QX = Nut Self-Locking Hexagon, non-metallic locking feature, .250-20 UNC-3B thread, Grade 5 steel, Zinc-Nickel, Type K16 plated.

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TABLE I. Dash numbers and dimensions.

Dash No.	Thread Size	A		B	C		D
		Width Across Flats		Corners	Thickness		
		MAX	MIN	REF	MAX	MIN	REF
00F	.060-80 UNF-3B	.111	.104	.116	.080	.055	.045
01C	.073-64 UNC-3B	.251	.243	.268	.153	.133	.081
01F	.073-72 UNF-3B	.251	.243	.268	.153	.133	.081
02C	.086-56 UNC-3B	.251	.243	.268	.153	.133	.081
02F	.086-64 UNF-3B	.251	.243	.268	.153	.133	.081
03C	.099-48 UNC-3B	.251	.243	.268	.153	.133	.081
03F	.099-56 UNF-3B	.251	.243	.268	.153	.133	.081
04C	.112-40 UNC-3B	.251	.243	.268	.153	.133	.081
04F	.112-48 UNF-3B	.251	.243	.268	.153	.133	.081
05C	.125-40 UNC-3B	.251	.243	.268	.153	.133	.081
05F	.125-44 UNF-3B	.251	.243	.268	.153	.133	.081
06C	.138-32 UNC-3B	.313	.305	.339	.188	.168	.103
06F	.138-40 UNF-3B	.313	.305	.339	.188	.168	.103
08C	.164-32 UNC-3B	.345	.336	.374	.239	.219	.140
08F	.164-36 UNF-3B	.345	.336	.374	.239	.219	.140
010C	.190-24 UNC-3B	.376	.367	.410	.249	.229	.140
010F	.190-32 UNF-3B	.376	.367	.410	.249	.229	.140
012C	.216-24 UNC-3B	.439	.430	.482	.328	.298	.225
012F	.216-28 UNF-3B	.439	.430	.482	.328	.298	.225
4C	.250-20 UNC-3B	.439	.428	.482	.328	.298	.225
4F	.250-28 UNF-3B	.439	.428	.482	.328	.298	.225
5C	.312-18 UNC-3B	.502	.489	.552	.359	.329	.250
5F	.312-24 UNF-3B	.502	.489	.552	.359	.329	.250
6C	.375-16 UNC-3B	.564	.551	.622	.468	.438	.335
6F	.375-24 UNF-3B	.564	.551	.622	.468	.438	.335
7C	.437-14 UNC-3B	.627	.616	.698	.468	.438	.324
7F	.437-20 UNF-3B	.627	.616	.698	.468	.438	.324
8C	.500-13 UNC-3B	.752	.736	.837	.609	.579	.464
8F	.500-20 UNF-3B	.752	.736	.837	.609	.579	.464
9C	.562-12 UNC-3B	.877	.861	.978	.656	.626	.469
9F	.562-18 UNF-3B	.877	.861	.978	.656	.626	.469
10C	.625-11 UNC-3B	.940	.922	1.051	.765	.735	.593
10F	.625-18 UNF-3B	.940	.922	1.051	.765	.735	.593
12C	.750-10 UNC-3B	1.064	1.052	1.191	.890	.860	.742
12F	.750-16 UNF-3B	1.064	1.052	1.191	.890	.860	.742
14C	.875-9UNC-3B	1.252	1.239	1.403	.999	.969	.790
14F	.875-14 UNF-3B	1.252	1.239	1.403	.999	.969	.790
16C	1.000-8UNC-3B	1.440	1.427	1.615	1.078	1.016	.825
16F	1.000-12 UNF-3B	1.440	1.427	1.615	1.078	1.016	.825
18C	1.125-7UNC-3B	1.627	1.614	1.826	1.203	1.141	.930
18F	1.125-12 UNF-3B	1.627	1.614	1.826	1.203	1.141	.930
20C	1.250-7UNC-3B	1.815	1.801	2.038	1.422	1.360	1.125
20F	1.250-12 UNF-3B	1.815	1.801	2.038	1.422	1.360	1.125
22C	1.375-6UNC-3B	2.008	1.973	2.232	1.609	1.547	1.282
22F	1.375-12 UNF-3B	2.008	1.973	2.232	1.609	1.547	1.282
24C	1.500-6UNC-3B	2.197	2.159	2.444	1.640	1.578	1.313
24F	1.500-12 UNF-3B	2.197	2.159	2.444	1.640	1.578	1.313

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TABLE II. Material coding.

MATERIAL CODE	
Steel Grade 5	G5
Steel Grade 8	G8
CRES 303*	S3
CRES 316	S6
Brass	BB
Aluminum	AA

* Or any Group 1 alloy in ASTM F594

TABLE III. Finish coding.

Finish	Code	Specification
Cadmium /1	C	SAE-AMS-QQ-P-416 Type II, Class 2 or 3
Zinc	Z	ASTM B633 Type II /2, Fe/Zn 8
Zinc (hex chrome-free)	F	ASTM F1941/F1941M Fe/Zn 8AN or 8BN, sealant optional
Black Oxide	B	MIL-DTL-13924, Class 1 or MIL-DTL-495 /3 (for brass)
Nickel	N	SAE-AMS-QQ-N-290, Class 2
Zinc-Nickel (Type K16)	QX	MIL-PRF-32647, Type K16
Zinc-Nickel (Type K20)	QW	MIL-PRF-32647, Type K20 /2
Anodized	no code (aluminum only)	MIL-PRF-8625, Type IC, II or IIB, Class 1 (clear, no color); sealers shall not contain hexavalent chromium compounds
Passivate	no code (stainless only)	SAE-AMS 2700, Method 1, Type 6 or 7, or Method 2, Class 1 (for CRES 303 or 303SE, Method 2 only)
Uncoated	no code	with dry to touch lubricant (for steel), or uncoated bright finish (for brass), hexavalent chromium prohibited

1/ Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL- PRF-32647, Type K16 is a suitable replacement with design authority approval.
2/ Type II coating per ASTM B633 contains hexavalent chromium which is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc with hexavalent chromium free passivation per ASTM F1941/F1941M Fe/Zn 8AN or 8BN, sealant optional or Zinc-Nickel plate per MIL- PRF-32647, Type K20 are suitable replacements with design authority approval.
3/ Hexavalent chromium compounds shall be prohibited.

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NOTES:

1. Dimensions in inches, unless specified, tolerance: ±.016.
2. Interpret drawing in accordance with ASME Y14.5.
3. Break all Sharp edged .003 to .005 and remove all burrs and slivers.
4. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.
5. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.
6. Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - OS
Air Force - 71

Preparing activity:
Army - AT

(Project 5310-2025-003)

Review activities:
Army - AV, CR4, EA, GL, MI
Navy - AS

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